

Dependencies

1. Ensure you have a GitHub account - if you don't set one up
2. Give your GitHub username or email address associated to your GitHub to Josh/Jemma/Harry
3. Go to <https://portal.glazer.ee> and check you can login
4. Go to <https://pinpoint.humava.io> and check you can login



UK Government

 TAG International

Africa
Cyber
Programme

CYBER INVESTIGATIONS

Open-Source Intelligence | Dark Web | Cryptocurrency

4-Day Course (15 Sep - 18 Sep 2026)

PARTNERING TOWARDS A GLOBAL
CYBER VISION

Aim

To develop an analyst's ability to investigate artefacts collected during an incident and use investigative techniques to build a comprehensive, evidence-led intelligence picture

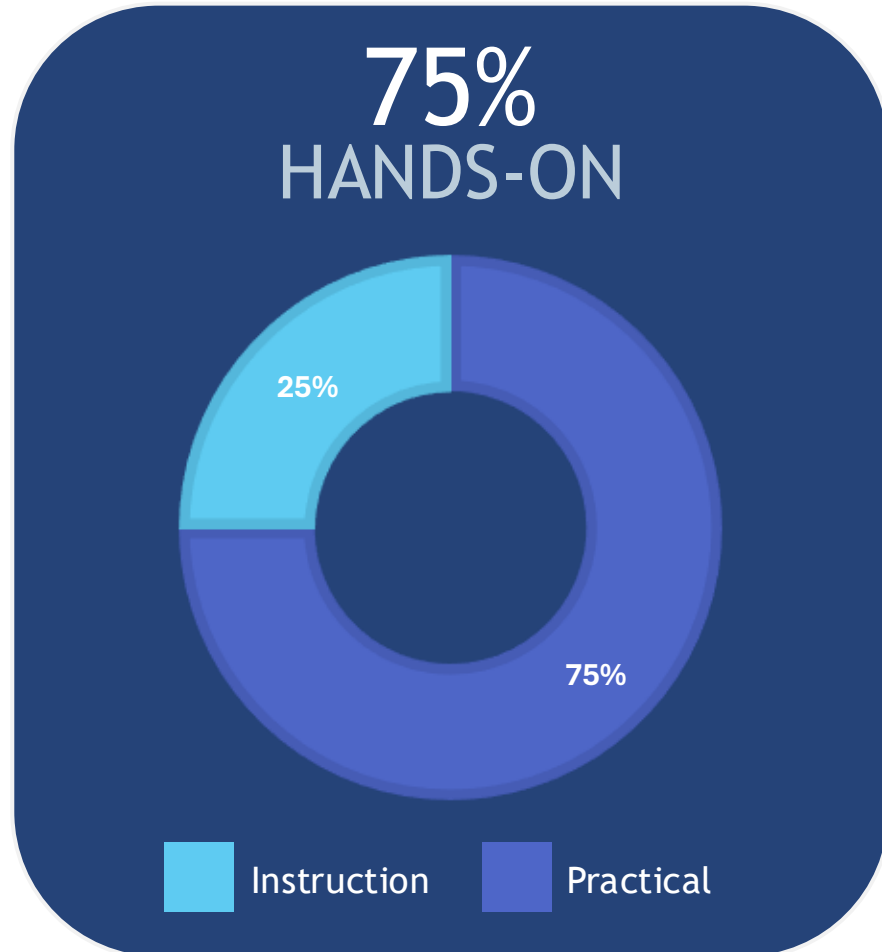
Objectives

- **Analyse** artefacts to identify indicators and relationships
- **Pivot** from identified information to develop further leads
- Use **tools** and **data sources** to support analysis
- **Apply** OSINT, dark web, and cryptocurrency investigative techniques
- Develop a **structured**, **repeatable**, and **auditable** investigative process
- Record and present findings with a **clear evidential and analytical rationale**

Purpose

To prepare participants for OPERATION BROKEN LINK and develop investigative skills that can be applied to real world scenarios

Course Delivery



End-to-End Case

Investigate yourself in a purpose-built environment



Group Work

Debriefs & case collaboration



Instructor Presentations

Tool walkthroughs, theory fundamentals & case debriefs



Homework

You may have to look over what has been discovered

Course Outline

DAY 1: Detection, Triage and Building an Investigation

DAY 2: Host Investigation, Impact and Cryptocurrency

DAY 3: Infrastructure, OSINT and Threat Actor Development

DAY 4: Extortion Assessment, Personas and Engagement

GUARDRAILS

SAFE, CONTROLLED & AUTHORISED INVESTIGATION

Guardrails

- Training environment only
- Stay within the exercise
- Maintain separation
- Protect your identity
- Document your decisions
- Escalate if unsure - we are here to guide and support you

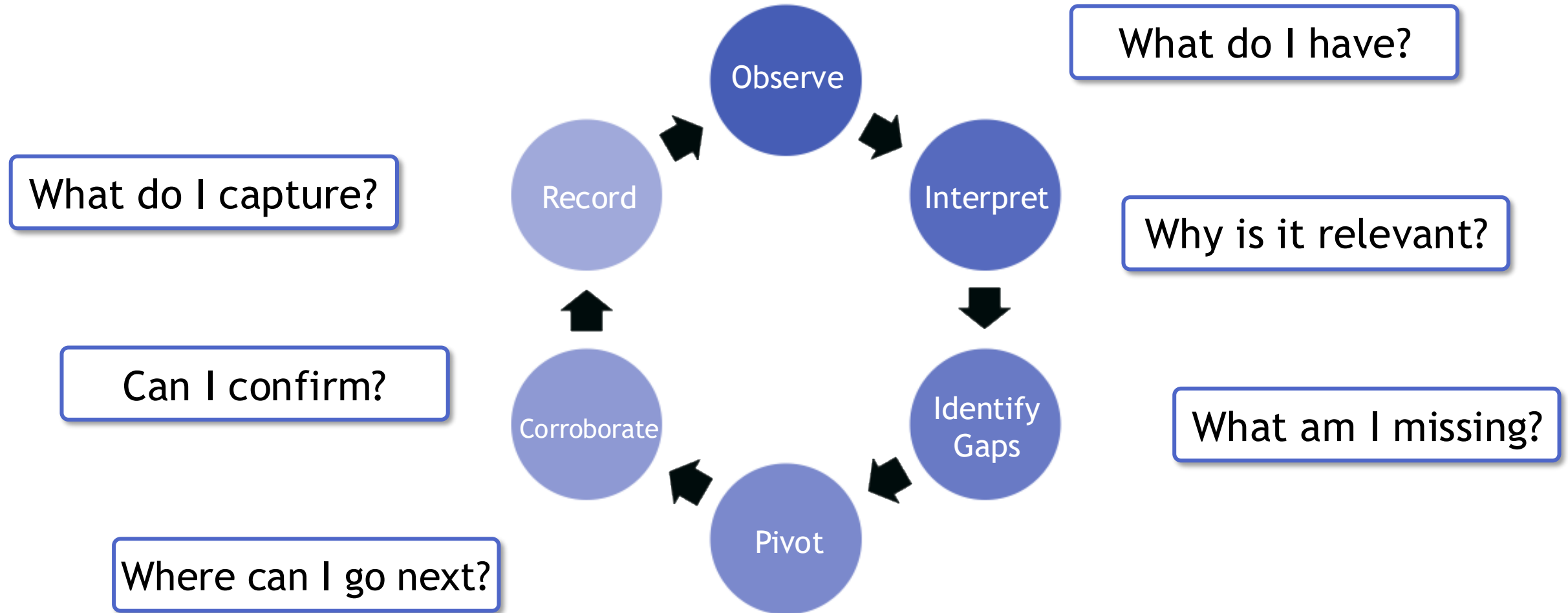


DAY 1

Detection, Triage and Building an Investigation

INVESTIGATIVE MODEL

Analytical Process



Working an Investigation

Start with a
question

Build the picture as
you go

Maintain a
chronology

Keep a record of
your investigative
trail

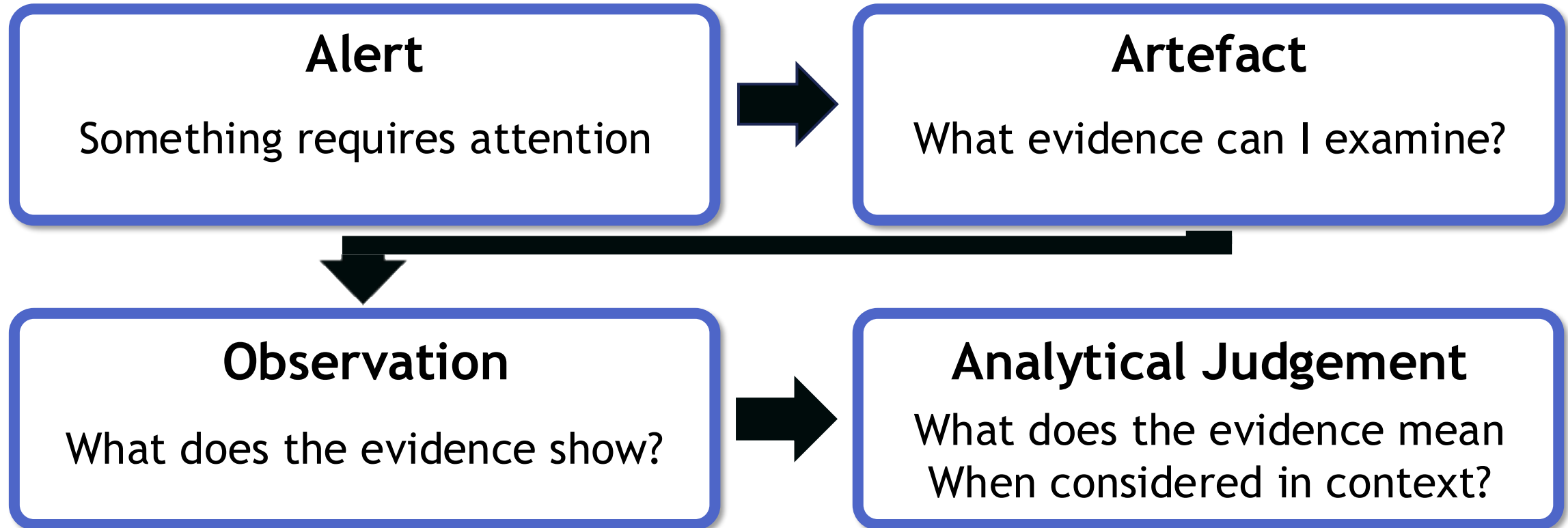
Be willing to change
initial judgements

Accept unresolved
questions

Know what you are trying to establish

DETECTION & TRIAGE

Investigative Information



The discipline is knowing when you move from observing to assessing

Indicators of Compromise (IOC)

Observable information that may indicate malicious or suspicious activity

EXAMPLES:

Domain/
URL

IP
address

File
name

File
hash

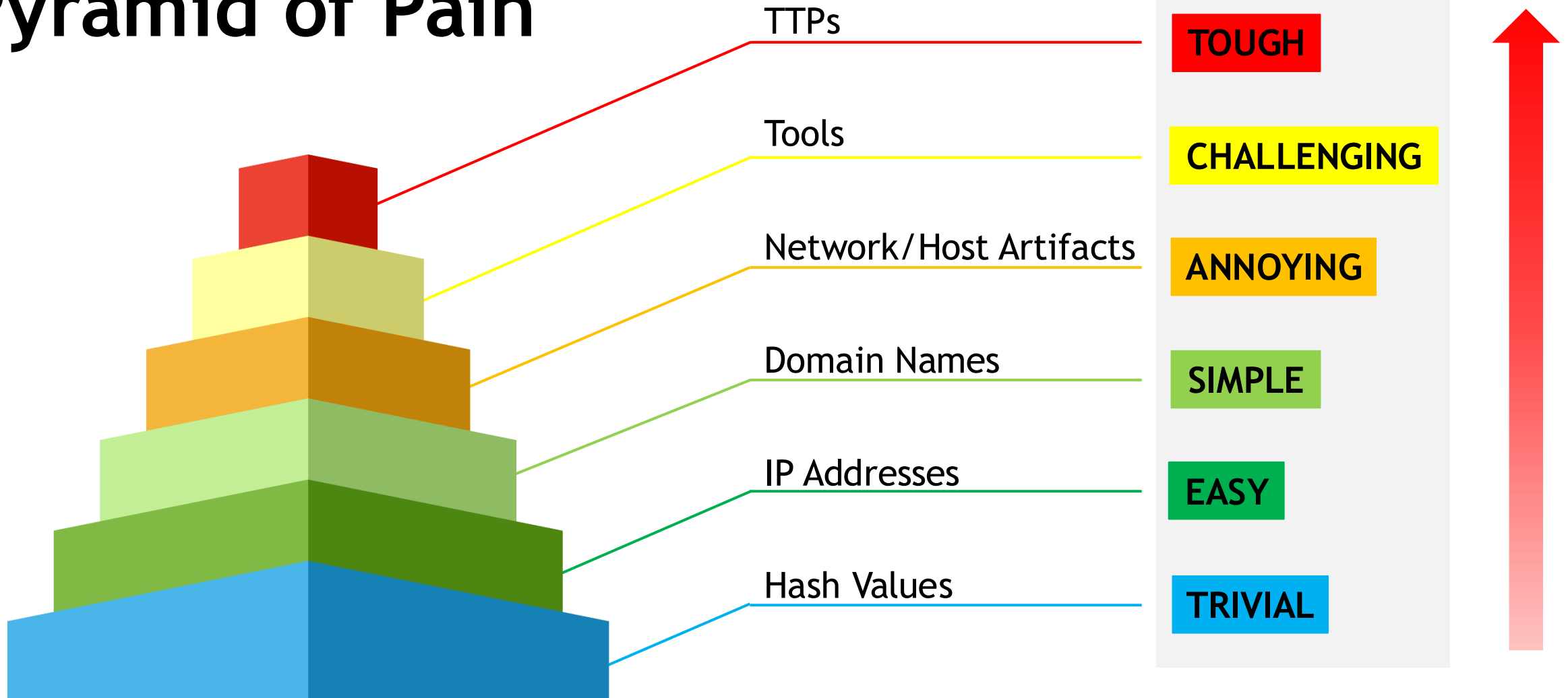
Email
address

Process

Alias

An IOC is a lead - not proof of malicious activity

Pyramid of Pain



PINPOINT FUNDAMENTALS

TOOL WALKTHROUGH

Recording an Investigation

WHY DO WE RECORD INVESTIGATIONS?



Accountability

We must be able to justify our decisions



Legal

The investigation might be used in the future



Continuity

Another analyst can take over if needed



Reproducibility

The conclusion you have reached can be understood



Traceability

Findings can be traced back to their source



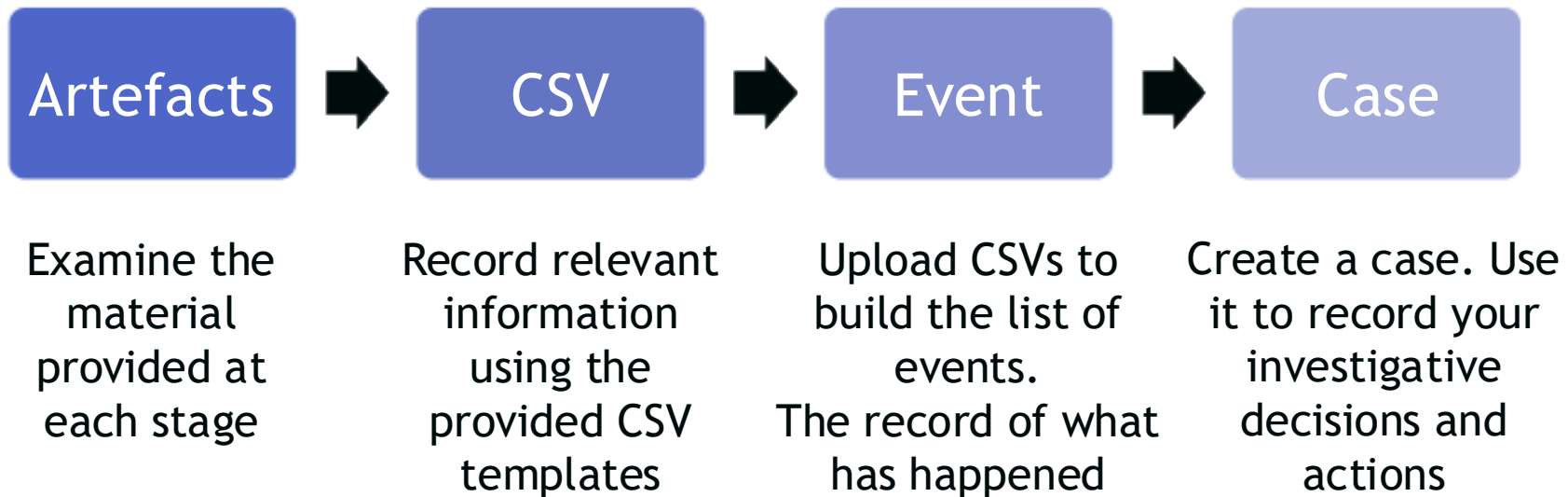
Lessons Learned

The record shows what worked, what didn't & gaps

If only you know what has happened, it's a hobby not an investigation

PinPoint

➤ Your investigative workspace



Events record the detail. The case records your actions

OPERATIONAL SECURITY (OPSEC)

Operational Security (OPSEC)

Online investigations can expose the investigator as well as the subject

IDENTITY
EXPOSURE

TECHNICAL
EXPOSURE

INVESTIGATIVE
EXPOSURE

OPSEC is controlling that exposure

Glazer

- Designed as a safe environment to conduct investigations
- Secure, isolated web browser
- Protects investigator identity
- Access surface and dark web



Conduct all investigative activity within Glazer

OPSEC Best Practice

APPROVED ENVIRONMENT

Use Glazer the supplied investigative environment

IDENTITY SEPARATION

Never use personal or organisational identities

MANAGED PERSONA

Use only approved exercise personas

NO UN-CONTROLLED INTERACTION

Do not register, message, purchase, download or otherwise interact unless authorised

RECORD DECISIONS

Document what you did, why you did it and any approval required

Tool Distinction



Training
Environment



Where you conduct your investigation



Case & SOAR
Management System



Where you record your investigation

ARTEFACT FUNDAMENTALS

Part 1

Alerts

An alert is normally the first thing we see in an incident

Something unusual has happened



Security systems kick-in



We could have 1 or multiple alerts

It is our responsibility as cyber security practitioners to **investigate** and **assess** if this activity is malicious

Often with incomplete information

```
alert_id: "BX-2026-0820-0117"
detection_time: "2026-08-20T09:14:52Z"
severity: "High"
▼ endpoint:
  hostname: "MAN-OPS-WS017"
  site: "Manchester Distribution Hub"
  os: "Windows 11 Enterprise"
▼ user:
  account: "j.harding"
  department: "Fleet Operations"
▼ process_chain:
  observed_process: "mshta.exe"
  command_line: "mshta.exe hxxps://portal-update.brightaxle-training.example/loader.hta"
  parent_process: "WINWORD.EXE"
  parent_document: "Vehicle_Maintenance_Schedule_Q3.docm"
▼ network:
  remote_domain: "portal-update.brightaxle-training.example"
  remote_ip: "198.51.100.27"
  destination_port: 443
  protocol: "HTTPS"
▼ file_artifact:
  filename: "training_placeholder.txt"
  sha256: "41f15571d687048950ab707384adcf03218cee19dbb068a67612e7f67e676766"
  note: "Harmless training-only file hash, generated for exercise purposes."
detection_summary: "Macro-enabled document spawned mshta.exe, which initiated an outbound HTTPS connection to a newly-registered external domain. No further lateral movement observed at time of alert."
recommended_action: "Isolate endpoint, preserve process tree, and review recipient's inbox for the originating email."
```

Email Threats

How does an adversary initiate an attack via email?



Initial Access

Emails are easy to acquire, easy to weaponize and hard to recognise as a threat



Versatile

Attachments, Links, blackmail and business email compromise (BEC)



Evasion

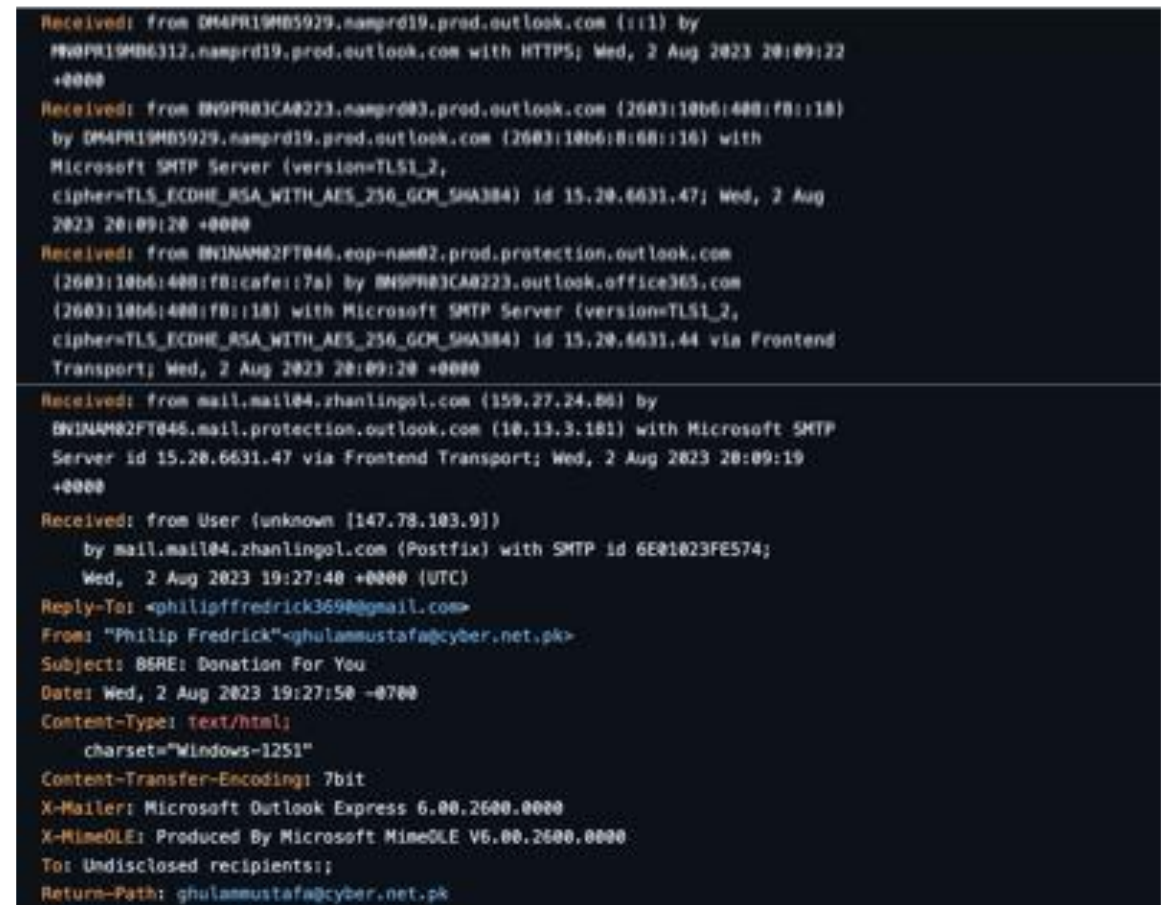
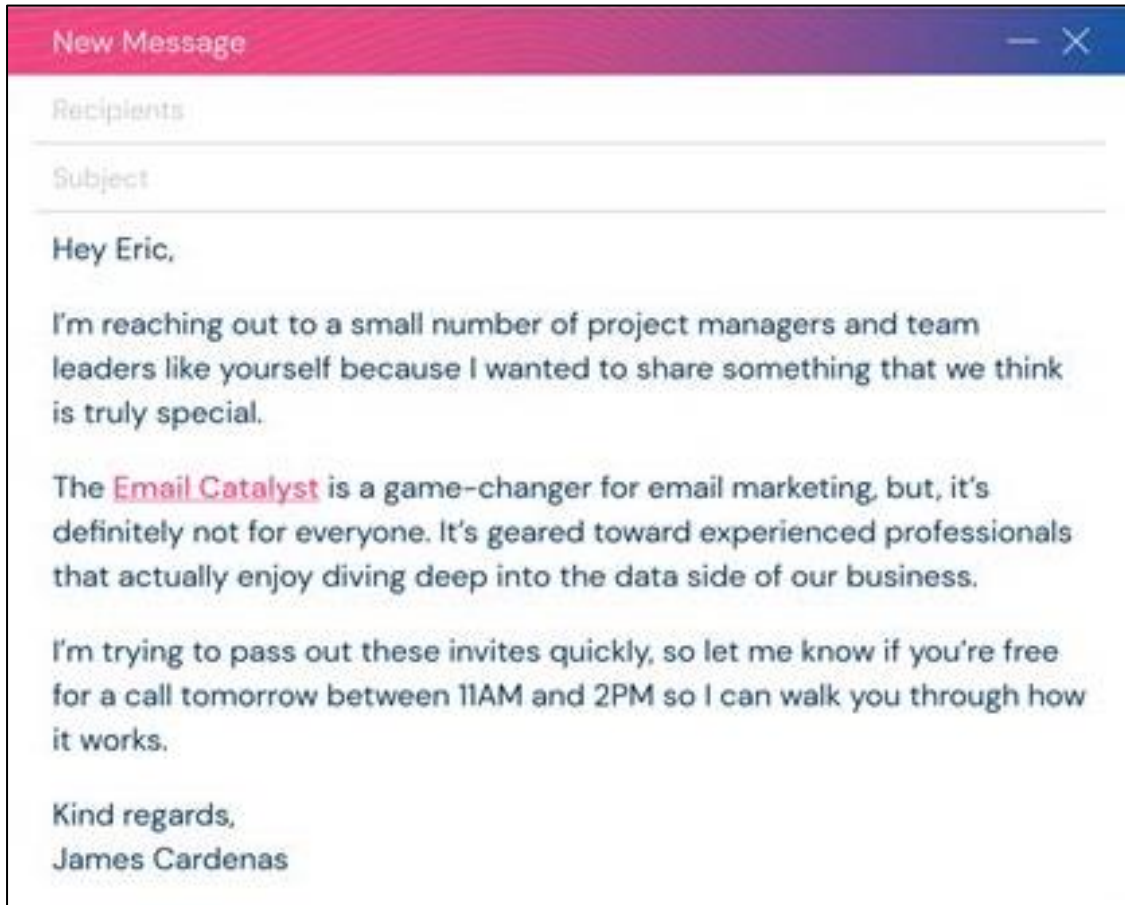
Easy to hide

SOCIAL ENGINEERING

When an attacker accomplishes malicious activities by tricking the victim into performing human interactions

It is crucial we understand email threats to identify & prevent attacks

Analysing Email Headers



Received: from D64PR19MB5929.namprd19.prod.outlook.com (::1) by
F80PR19MB6312.namprd19.prod.outlook.com with HTTPS; Wed, 2 Aug 2023 20:09:22
+0000

Received: from B09PR03CA0223.namprd03.prod.outlook.com (2603:10b6:400:f0::18)
by D64PR19MB5929.namprd19.prod.outlook.com (2603:10b6:8:68::16) with
Microsoft SMTP Server (version=TLS1_2,
cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) id 15.20.6631.47; Wed, 2 Aug
2023 20:09:20 +0000

Received: from B01NAM02FT046.eop-nam02.prod.protection.outlook.com
(2603:10b6:400:f0:cafe::7a) by B09PR03CA0223.outlook.office365.com
(2603:10b6:400:f0::18) with Microsoft SMTP Server (version=TLS1_2,
cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) id 15.20.6631.44 via Frontend
Transport; Wed, 2 Aug 2023 20:09:20 +0000

Received: from mail.mail04.zhanlingol.com (159.27.24.86) by
B01NAM02FT046.mail.protection.outlook.com (10.13.3.181) with Microsoft SMTP
Server id 15.20.6631.47 via Frontend Transport; Wed, 2 Aug 2023 20:09:19
+0000

Received: from User (unknown [147.78.103.9])
by mail.mail04.zhanlingol.com (Postfix) with SMTP id 6E01023FE574;
Wed, 2 Aug 2023 19:27:40 +0000 (UTC)

Reply-To: <philipffredrick369@gmail.com>

From: "Philip Fredrick" <ghulamustafa@cyber.net.pk>

Subject: BERE: Donation For You

Date: Wed, 2 Aug 2023 19:27:50 -0700

Content-Type: text/html;
charset="Windows-1251"

Content-Transfer-Encoding: 7bit

X-Mailer: Microsoft Outlook Express 6.00.2600.0000

X-MimeOLE: Produced By Microsoft MimeOLE V6.00.2600.0000

To: Undisclosed recipients;

Return-Path: ghulamustafa@cyber.net.pk

THE CASE

What has happened?

1

THE ORGANISATION

Regional IP backbone provider operating across Nigeria



2

THE SITUATION

Potential malicious activity has been identified within the corporate environment

3

CURRENT POSITION

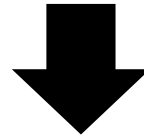
The activity has been referred to your team for investigation

YOUR TASK

Establish what has happened, determine the extent of any compromise, & follow the evidence

Evidence Location

Stage 1
Artefacts



<https://github.com/NG-OSINT/Cyber-Investigation-Artefacts>

Record your investigation using your PinPoint records